Conformal $U(1)_D$ Dark-Sector First-Order Phase Transition and the NANOGrav Signal

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Abstract

We report the DarkAgents analysis of the prompt: "Can a conformal U(1) BSM extension with a dark scalar, a dark gauge bosons and a dark fermion explain the NANOGrav signal with a cosmological FOPT?" The conditional answer is yes at the level computed here: the validated conformal Abelian dark sector has viable first-order phase-transition points, a benchmark scan contains points lying inside all 14 PTA violin windows, and a PTArcade run on NANOGrav 15-year data finds a posterior region near $v_D \simeq 1$ GeV and $g_D \simeq 0.29$. This is not yet a full exclusion-safe cosmological model. The result depends on an approximate semianalytic FOPT backend, a DBF gravitational-wave template used outside strict validity for strongly supercooled points, exact-zero portal assumptions, and missing BBN/CMB, relic-density, thermal-history, RG-validity and completion calculations.

1 Model

The validated model is the Standard Model plus a dark $U(1)_D$. All new fields are SM singlets. The dark scalar Φ has $U(1)_D$ charge 2, the dark gauge boson is X_μ, and the fermion sector contains two left-handed Weyl fields ψ_+ and ψ_- with charges $+1$ and -1. This charge assignment permits Majorana-type Yukawa terms after Φ gets a Coleman-Weinberg vev and is anomaly-safe for the Abelian dark gauge symmetry.

The dark-sector Lagrangian terms used by the validated handoff are

$$\mathcal{L}_D \supset -\frac{1}{4}X_{\mu\nu}X^{\mu\nu} + |D_\mu\Phi|^2 + i\psi_+^\dagger\bar{\sigma}^\mu D_\mu\psi_+ + i\psi_-^\dagger\bar{\sigma}^\mu D_\mu\psi_- - \frac{1}{2}y_D\Phi^\dagger\psi_+\psi_+ - \frac{1}{2}y_D\Phi\psi_-\psi_- + \text{h.c.} - \lambda_D(\Phi^\dagger\Phi)^2. \tag{1}$$

The symmetry-allowed Higgs portal $\lambda_{HP}(H^\dagger H)(\Phi^\dagger\Phi)$ and hypercharge kinetic mixing $\epsilon B_{\mu\nu}X^{\mu\nu}/2$ are fixed to zero in the backend. The independent scan parameters are v_D, g_D, and y_D. The dependent quantities are

$$g_b = 2g_D, \qquad g_f = \frac{y_D}{\sqrt{2}}, \qquad \beta_\lambda = \frac{3g_b^4 - 4g_f^4}{8\pi^2}, \tag{2}$$

with Coleman-Weinberg scalar mass $m_\chi^2 = \beta_\lambda v_D^2$. The field-dependent masses used for the backend are

$$m_X^2(\chi) = 4g_D^2\chi^2, \qquad m_\psi^2(\chi) = \frac{y_D^2}{2}\chi^2, \tag{3}$$

where the fermion entry represents two degenerate Majorana Weyl mass eigenstates with total fermionic degree count 4. The explicit thermal-mass formulae in the handoff are documentation placeholders, not direct semianalytic backend inputs.

2 Pipeline Status

Stage	Status	Interpretation
Proposal	warning	Usable conformal $U(1)_D$ model proposal; no numerical FOPT/PTA claim.
Preliminary literature	ok	Relevant conformal dark-sector PTA literature and benchmark seeds identified.
Critique/model validation	warning	No fatal model red flags; exact-zero portals, $\beta_\lambda > 0$, and backend caveats remain.
FOPT	ok	Semianalytic scan executed.
PTA/PTArcade	warning	Usable benchmark and PTArcade outputs with DBF-template and MAP-boundary warnings.
Constraints	warning	No non-PTA published constraint applied as an exclusion; several require new observables.
Prior audit	warning	Highest-severity unresolved issues are thermal history, portal assumptions, supercooling completion, and physical posterior masking.

3 Literature Context

The preliminary literature search found that the broad model class is not new: conformal or nearly conformal dark sectors have already been studied as PTA/NANOGrav FOPT explanations. The exact backend limit used here, with $q_\Phi = 2$, an anomaly-safe Weyl pair, $\epsilon = 0$, and $\lambda_{HP} = 0$, is a variant of that class rather than a fully distinct unexplored mechanism.

The closest verified paper is Balan, Bringmann, Kahlhoefer, Matuszak and Tasillo, "Sub-GeV dark matter and nano-Hertz gravitational waves from a classically conformal dark sector," arXiv:2502.19478. It contains a conformal hidden $U(1)'$, dark photon, dark Higgs, stable fermion, Yukawa coupling, and PTA/NANOGrav-relevant FOPT. Differences include charge normalization and nonzero portal/kinetic-mixing assumptions. Borah, Dasgupta and Kang, arXiv:2105.01007, studied a dark $U(1)_D$ FOPT in light of NANOGrav 12.5-year data, but with negligible dark Yukawa in the FOPT analysis. Goncalves, Marfatia, Morais and Pasechnik, arXiv:2501.11619, studied supercooled conformal dark $U(1)'$ transitions with PTArcade likelihoods. Fujikura, Girmohanta, Nakai and Suzuki, arXiv:2306.17086, is relevant to conformal dark PTA signals but uses a different confining mechanism. Hashino, Kanemura, Takahashi and Tanaka provide Abelian conformal phase-transition context through an INSPIRE-verified record, but were not used as a direct target-model benchmark.

4 FOPT Results

The FOPT agent used the local semianalytic conformal single-field backend. It scanned 791 total points: 416 were viable, 307 failed numerically, and 68 were excluded or physically failed. Among viable points, 187 had estimated peak-frequency scale in the PTA band. The backend caveats are important: it uses a small-field/high-temperature polynomial approximation, approximate percolation treatment, and no RG running, portal coupling, kinetic mixing, or full daisy-resummed finite-temperature potential.

Point	v_D [GeV]	g_D	y_D	α	β/H	$T_{\rm reh}$ [GeV]
extended_low_v0.002_g0.5_y0.5	0.002	0.50	0.50	0.804	253.7	3.53×10^{-4}
extended_broad_v0.08367_g0.25_y0.25	0.0837	0.25	0.25	2.04×10^{16}	13.0	6.68×10^{-3}
extended_low_v0.01108_g0.38_y0.5	0.0111	0.38	0.50	83.7	80.0	1.19×10^{-3}

The quantity $f_{\rm peak,est}$ in the handoff is an order-of-magnitude redshifted scale guide used to seed PTA analysis, not a true spectral peak.

5 PTA and PTArcade Results

The PTA agent used a DBF spectrum template. In the benchmark comparison, 19 viable points were found by the PTA scoring procedure and the best points scored 14/14, meaning their spectra fell inside all 14 tabulated PTA violin windows. The top point has $v_D = 0.7653$ GeV, $g_D = 0.35$, $y_D = 0.9$, $\alpha = 4.22 \times 10^{11}$, $\beta/H = 17.52$, $T_\star = 0.0469$ GeV, peak frequency 1.14×10^{-8} Hz, and peak $h^2\Omega_{\rm GW} = 4.80 \times 10^{-9}$. This is strong evidence of spectral compatibility with the PTA windows, but the strongest points have extremely large α, so the DBF template validity warning is central.

The PTArcade run used NANOGrav 15-year data in ceffyl mode. The priors were log-uniform $v_D \in [10^{-3.5}, 10]$ GeV, uniform $g_D \in [0.15, 1.0]$, and uniform $y_D \in [0, 0.95]$. The stored chains were finite; 10001 rows were stored after a completed $N_{\rm samples} = 100000$ run. The posterior estimates are

$$v_D = 0.960 \text{ GeV} \quad (\sigma_{\log_{10} v_D} = 0.182), \tag{4}$$

$$g_D = 0.286 \pm 0.026, \qquad y_D = 0.405 \pm 0.255. \tag{5}$$

The 68% intervals are $v_D = 0.652$–1.395 GeV, $g_D = 0.260$–0.315, and $y_D = 0.119$–0.705. The MAP point is $v_D = 1.176$ GeV, $g_D = 0.262$, $y_D = 0$. Since y_D is at the physical lower boundary, the MAP must not be interpreted as an interior preference for a vanishing Yukawa.

6 Constraints

The constraint handoff considered 12 constraints. None was directly testable as a published exclusion with the present observables. Two were approximately testable, two require a recast, four require new backend calculations, and four were not applicable in the exact-zero-portal backend limit.

ID	Constraint	Status	Reason
C01	BBN expansion/dark radiation	requires new backend	Needs dark-sector temperature ratio, decoupling, abundances, and lifetimes.
C03	CMB $\Delta N_{\rm eff}$	requires new backend	Radiation density at BBN/recombination was not computed.
C04	CMB energy injection	recast needed	Requires abundance, annihilation/decay channels, and SM-visible final states.

C05	Thermalization/decoupling	requires new backend	Zero portals leave the dark-sector temperature history unspecified.
C06–C08	Dark photon and scalar laboratory limits	not applicable	Published limits require nonzero kinetic mixing or Higgs portal; backend fixes both to zero.
C09	Dark fermion relic abundance	requires new backend	Stability and Boltzmann abundance are not established.
C10	Self-interactions/structure formation	recast needed	Requires DM identity, relic fraction, and velocity-dependent cross sections.
C11	Supercooling completion	approximate	Large-α points need stronger completion and wall-velocity diagnostics.
C12	Perturbativity/CW validity	approximate	Couplings are perturbative, but rectangular 68% posterior corners include $\beta_\lambda < 0$; RG running was not computed.

The preferred 68% PTArcade region maps to $m_X = 0.340$–0.878 GeV and $m_\psi = 0.055$–0.695 GeV under $m_X = 2 g_D v_D$ and $m_\psi = y_D v_D/\sqrt{2}$. For $\beta_\lambda > 0$ corners, the estimated scalar mass range is $m_\chi \simeq 0.034$–0.108 GeV. Because these ranges use independent rectangular intervals rather than a masked posterior covariance, they should be treated as approximate.

7 Prior and Assumption Audit

The highest-severity uncontrolled issues are:

ID	Category	Issue	Severity
A01	portal	Exact $\epsilon = 0$ removes visible dark-photon production, decay, and thermalization channels without a protective argument.	8
A02	portal	Exact $\lambda_{HP} = 0$ removes scalar mixing and SM thermal contact without a protective argument.	8
A05	physical mask	The rectangular PTArcade 68% region contains $\beta_\lambda < 0$ corners.	8
A07	percolation	Strongly supercooled points need robust completion, reheating, and wall-velocity diagnostics.	8
A09	thermal history	No BBN/CMB/$\Delta N_{\rm eff}$ thermal-history calculation is present.	8
A08	GW template	The DBF template is the closest supported template, not a strict template for all very strong transitions.	7
A10	dark matter	Fermion stability and relic abundance are unresolved.	7
A12	RG validity	No RG, Landau-pole, or loop-stability audit was performed.	7

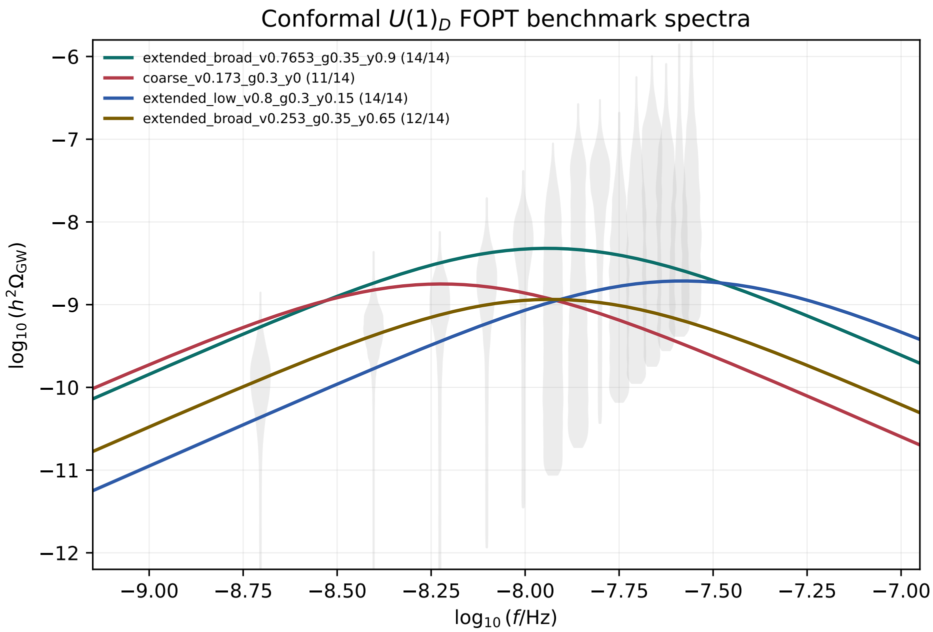

Figure 1: Benchmark DBF spectrum compared with the PTA violin windows.

The highest-priority followups are a $\beta_\lambda > 0$-masked posterior summary, a dark-sector thermal-history module, a full completion/percolation cross-check of the preferred and 14/14 points, justification or relaxation of exact-zero portals, and a dark fermion stability/relic-density calculation.

8 Conclusion

Within the implemented workflow, the model can conditionally explain the NANOGrav signal through a cosmological first-order phase transition. This statement is supported by a successful semianalytic FOPT scan, PTA benchmark spectra that include 14/14 violin-window matches, and a finite PTArcade chain whose posterior lies near $v_D \sim 1$ GeV and $g_D \sim 0.3$. The result should be read as positive evidence for spectral compatibility, not as a completed cosmological viability proof.

The unresolved caveats are not cosmetic. Exact-zero portals control whether visible-sector laboratory, BBN and CMB constraints apply; the dark-sector temperature history is absent; the strongest PTA-compatible points are supercooled and require completion diagnostics beyond the current approximation; the DBF template is only a closest supported template; the posterior must be masked to enforce $\beta_\lambda > 0$; and the dark fermion has not been promoted to a fully calculated relic. No non-PTA constraint currently excludes the reported preferred region, but this is a non-exclusion due to missing observables and backend scope, not a proof that the region is safe.

References

1. Debasish Borah, Arnab Dasgupta, Sin Kyu Kang, "Gravitational waves from a dark $U(1)_D$ phase transition in the light of NANOGrav 12.5 yr data," arXiv:2105.01007, Phys. Rev. D 104, 063501 (2021).

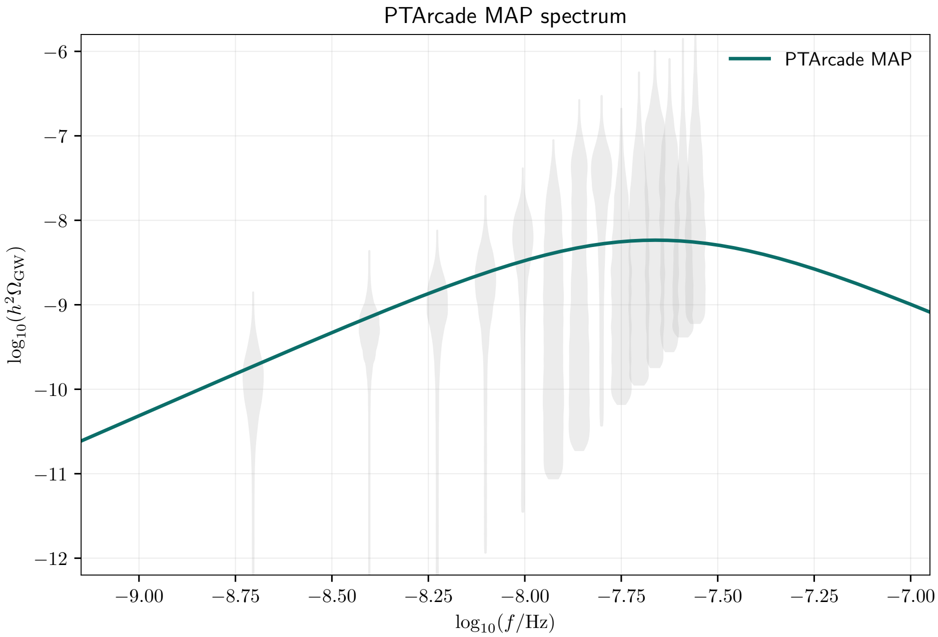

Figure 2: PTArcade MAP spectrum generated from the final chain summary.

2. Joao Goncalves, Danny Marfatia, Antonio P. Morais, Roman Pasechnik, "Supercooled phase transitions in conformal dark sectors explain NANOGrav data," arXiv:2501.11619, Phys. Lett. B 869, 139829 (2025).

3. Sowmiya Balan, Torsten Bringmann, Felix Kahlhoefer, Jonas Matuszak, Carlo Tasillo, "Sub-GeV dark matter and nano-Hertz gravitational waves from a classically conformal dark sector," arXiv:2502.19478, JCAP 08 (2025) 062.

4. Kohei Fujikura, Sudhakantha Girmohanta, Yuichiro Nakai, Motoo Suzuki, "NANOGrav Signal from a Dark Conformal Phase Transition," arXiv:2306.17086.

5. Katsuya Hashino, Shinya Kanemura, Yuta Takahashi, Masatoshi Tanaka, "Phase transition and vacuum stability in the classically conformal B-L model," INSPIRE record 1705442.

6. Marco Hufnagel, Kai Schmidt-Hoberg, Sebastian Wild, "BBN constraints on MeV-scale dark sectors. Part I. Sterile decays," arXiv:1712.03972.

7. Marco Hufnagel, Kai Schmidt-Hoberg, Sebastian Wild, "BBN constraints on MeV-scale dark sectors. Part II. Electromagnetic decays," arXiv:1808.09324.

8. Eleonora Di Valentino, Alessandro Melchiorri, Joseph Silk, "Cosmological constraints in extended parameter space from the Planck 2018 Legacy release," arXiv:1908.01391.

9. Torsten Bringmann, Felix Kahlhoefer, Kai Schmidt-Hoberg, Parampreet Walia, "Strong constraints on self-interacting dark matter with light mediators," arXiv:1612.00845.

10. M. Battaglieri et al., "US Cosmic Visions: New Ideas in Dark Matter 2017: Community Report," arXiv:1707.04591.

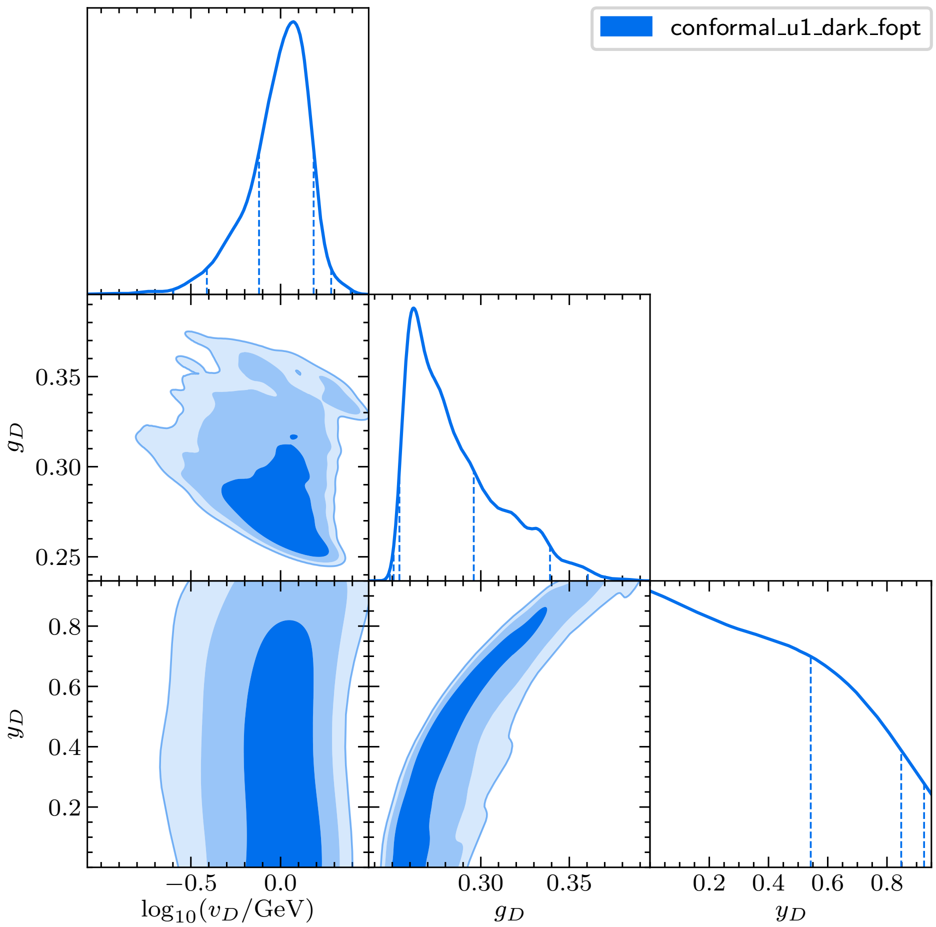

Figure 3: PTArcade posterior distributions for the model parameters.

11. Marco Fabbrichesi, Emidio Gabrielli, Gaia Lanfranchi, "The Dark Photon," arXiv:2005.01515.

12. Thomas Hambye, Laurent Vanderheyden, "Minimal self-interacting dark matter models with light mediator," arXiv:1912.11708.

13. Ryusuke Jinno, Thomas Konstandin, Hyeonseok Rubira, Jorinde van de Vis, "Supercool subtleties of cosmological phase transitions," arXiv:2212.07559.

14. Alexander Farzinnia, Hong-Jian He, Jing Ren, "Classically conformal B-L extended Standard Model and Higgs vacuum stability," arXiv:1205.2893. This citation is marked uncertain in the constraint handoff metadata and is not used as an exclusion claim here.